

Adroit Technologies

Protocol Driver

Name	JVA Ethernet Client Protocol Driver
DLL	D_JVA_ETH



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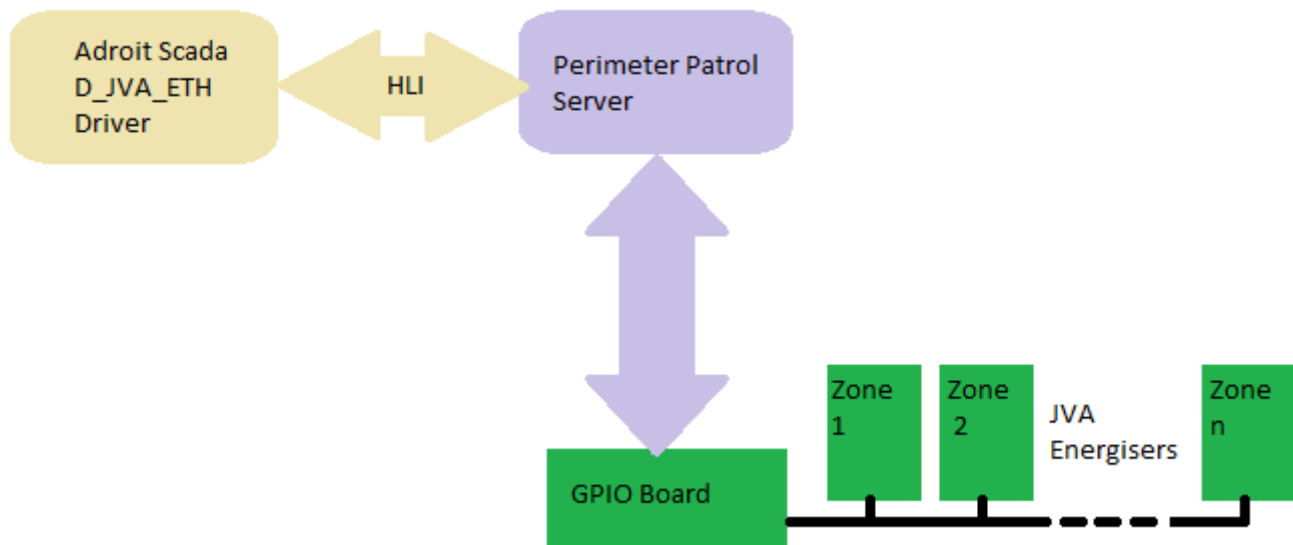
<https://adroittech.co.za/>

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1. Overview

HLI(High level interface) SDK based driver allows for communication between the Adroit Scada server and the Perimeter Patrol server.



Live data can be read in the form of discrete I/O communicated in the form of Boolean, Integer, Real/Analog , String/Var-String types.

Live Data can also be read in XML communicated to an Adroit IOT Agent.

Zone statuses can also be issued to the system in the form of an action string, such as "Disarm", "ArmHighPower" see table 4.2

A writeable bitwise output word if available is also available.

PLEASE NOTE

The HLI updates all I/O on the channel once every second. Scanning anyfaster than this will not affect this rate, scanning slower will just result in a slower retrieval from the internal store of changes within the HLI.

1.1. Supported Hardware

This driver has been tested on the following hardware:

GPIO board revision xxxx.

Z18 JVA energiser revision xxxx.

1.2. Supported Software

Perimeter Patrol version xxxx.

HLI SDK version 1.0.0.6015.

PP.HLITransport.dll (version 1.0.0.6015)

2. Wiring

Standard Ethernet interface provided between the GPIO board and the Perimeter Patrol server computer. The Adroit Scada computer also needs ethernet access to the Perimeter Patrol server computer.

The GPIO communication bus to the energisers as per JVA documentation.

See Appendix X for setup howto and the following documentation : x.

3. Device Configuration

3.1. Device Configuration

JVA HLI Ethernet TCP/IP Driver - PP1

☐ Enable secondary ☒ Primary ☐ Secondary

Primary connection parameters

☐ Enable shadow scanning from master to standby ☒ Enable standby reads
☐ Enable standby writes

Property	Value
Perimeter Patrol server details	
TCP/IP Address	10.10.10.58
TCP Port	17289
Authorization Key	*****
User Name	admin
User Password	****
Datascope details	
Debug level bitmask	480
Other details	
Stale threshold	5

Close

Enable secondary – Turn this option to have the driver maintain 2 channels of communication, the radio buttons **Primary** and **Secondary** can then be used to toggle between configurations.

The Perimeter Patrol HLI interface does provide access for more than 1 channel , so either parallel or shadow mode clustering and primary and secondary connections per server is available

3.1.1. Device details

Enable Secondary – (Default = OFF) Allows enabling of redundant MQTT Broker connections such that in case of a failure a switchover will occur.

Primary – Allows configuration setting of the Primary broker connection.

Secondary – Allows configuration setting of the Secondary broker connection.

Enable shadow scanning from master to standby – (Default = OFF) If enabled will stop any communication to the MQTT broker when the Agent server is in standby mode. By default, we assume parallel scanning.

Enable standby reads – (Default = ON) Allows reading from the MQTT broker when agent server is in standby mode.

Enable standby writes – (Default = OFF) Allows writing to the MQTT broker when agent server is in standby mode.

3.1.2. Perimeter Patrol server details

TCP/IP Address – The static IP Address of the Perimeter patrol server machine.

TCP Port – A valid port number is required for the connection to take place. Configured in the perimeter patrol system, see Appendix A for details.

Authorization key – An text string used as a seed for the encryption algorithm, used when communicating to the Perimeter Patrol system. Configured in the perimeter patrol system, see Appendix A for details.

User Name – A username required by Perimeter Patrol. Configured in the perimeter patrol system, see Appendix A for details.

User Password – A password required by Perimeter Patrol. Configured in the perimeter patrol system, see Appendix A for details.

3.1.3. Datascope details

Debug level bitmask – (Default = 480) This is a bitmask that enable extra debugging in the datascope output, do not leave it enabled during normal operation on high-speed drivers as it will slow down the system depending on speed requirements. In serial drivers this delay is negligible.

#define DEBUG_MASK_COMSRESOURCE_INFO	0x00000001
#define DEBUG_MASK_COMSRESOURCE_CRITICAL	0x00000002
#define DEBUG_MASK_PROTOCOL_ITEM_VALUES	0x00000004
#define DEBUG_MASK_PROTOCOL_ITEM_BAD_VALUES	0x00000008
#define DEBUG_MASK_NASA_POOLS	0x00000010
#define DEBUG_MASK_TIMING	0x00000020
#define DEBUG_MASK_PROTOCOL_RESPONSES	0x00000040
#define DEBUG_MASK_PROTOCOL_REQUESTS	0x00000080
#define DEBUG_MASK_PROTOCOL_READ_DESC	0x00000100
#define DEBUG_MASK_PROTOCOL_WRITE_DESC	0x00000200

Stale threshold – (Default = 0) Every nth occurrence of an unchanged item value will be sent to the master. If the value is different, the value will be sent regardless of this count. This count reduces unnecessary event within the Agent Server. This can be considered an integrity verification update.



INFORMATION

Remember, all NASA drivers one can stop and start the device to reload Debug level bitmask flags, and most other parameters in setup. If in doubt restart the agent server as mechanisms like PDU lengths might not be changeable during runtime.

4. Supported Addresses

The scan address in Adroit for the JVA system is as follows :

When scanning discrete I/O:

<Tagname> <Space> <Type-postfix>

e.g., **Zones.1.FenceAlarm BOOL**

When scanning to the IOT agent's rawValue slot:

<Tagname> <Space> <Type-postfix>

e.g., **Zones.1 XML**



PLEASE NOTE

IOT slots cannot have additional '.' Dot notations as these are used within Adroit system itself, therefor the xml sent from the driver will replace any '.' With '_'

The Address field when scanning however does allow for '.' Notation, And the dot will be displayed in the Description field for clarification purposes.

Example : IOT agent scanned its rawValue to "Zones.1" where a subsequent slot such as AlarmOnStartTime.0 will be renamed to AlarmOnStartTime_0.

Type-Postfixes :

Data type	Prefix	Example
Boolean	BOOL	Zones.1.FenceAlarm BOOL
Long	LONG	Zones.1.Voltage1 LONG
Double	DWORD	Zones.1.ComsFailTimer DWORD
Integer	INT	Zones.1.Outputs INT
Word	WORD	Zones.1.Outputs WORD
Real	REAL/FLOAT	Zones.1.Voltage1 REAL

String[80]	STRING	
VarString	VARSTRING	Zones.1.ArmState VARSTRING
XML/IOT	XML	Zones.1 XML

String User events <i>(sent to Zones.x.ArmState)</i>
ArmAllHighPower
ArmAllLowPower
ArmHighPower
ArmLowPower
Disarm
DisarmAll
ClearAlarmMemory
CloseAlarms
ControlOnSchedule
ControlOffSchedule
MuteAlarms

PLEASE NOTE

IOT agents are read only , the ArmState control string MUST be scanned to a String agent as an output enabled discrete address.

E.g Zones.x.ArmState where x in the Zone number.

Some User event strings affect all devices/zones on a channel, andy zone number can accept these strings



INFORMATION

Prefixes attempt to force the driver to pre-convert the I/O to various types and must be included.

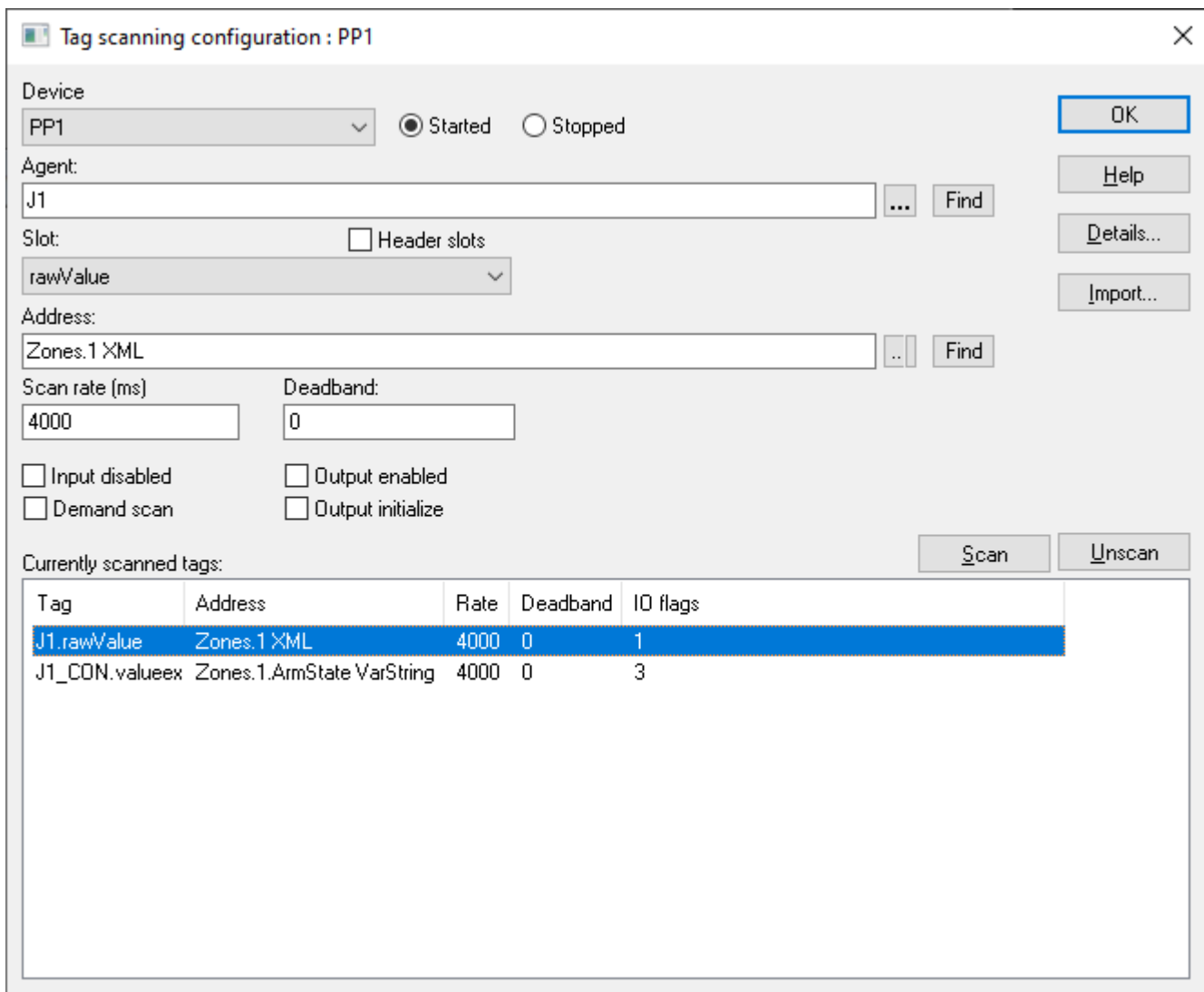
4.1. Example

PP1 is the channel / device name as configured.

J1 is the IOT agent associated with "Zone.1"

J1_CON is the writable string agent associated with "Zone.1"

The following screen shot shows the scanning configuration dialog:



Tag scanning configuration : PP1

Device: PP1 ☒ Started ☐ Stopped

Agent: J1 ... Find

Slot: ☐ Header slots
rawValue

Address: Zones.1 XML .. Find

Scan rate (ms): 4000 Deadband: 0

☐ Input disabled ☐ Output enabled
☐ Demand scan ☐ Output initialize

Currently scanned tags: Scan Unscan

Tag	Address	Rate	Deadband	IO flags
J1.rawValue	Zones.1 XML	4000	0	1
J1_CON.valueex	Zones.1.ArmState VarString	4000	0	3

The following screen shot shows the configuration view of the IOT agent at runtime:

Edit IOT agent : J1

Slot information:

Name	Description	Type	Routing agent type	Value
AcFail	Zones.1.AcFail	Boolean		0
LowBatt	Zones.1.LowBatt	Boolean		0
BadBatt	Zones.1.BadBatt	Boolean		0
Tamper	Zones.1.Tamper	Boolean		0
Fault	Zones.1.Fault	Boolean		0
FenceAlarm	Zones.1.FenceAlarm	Boolean		0
Gate	Zones.1.Gate	Boolean		0
OverVoltage1	Zones.1.OverVoltage1	Boolean		0
OverVoltage2	Zones.1.OverVoltage2	Boolean		0
OverVoltage3	Zones.1.OverVoltage3	Boolean		0
UnderVoltage1	Zones.1.UnderVoltage1	Boolean		0
UnderVoltage2	Zones.1.UnderVoltage2	Boolean		0
UnderVoltage3	Zones.1.UnderVoltage3	Boolean		0
GroundAlarm	Zones.1.GroundAlarm	Boolean		0
ControlOverride	Zones.1.ControlOverride	Boolean		0
Voltage1	Zones.1.Voltage1	Real		10
Voltage2	Zones.1.Voltage2	Real		10
ComsFailTimer	Zones.1.ComsFailTimer	Integer		0
ZoneIndex	Zones.1.ZoneIndex	Integer		1
HaveReceivedVoltage1	Zones.1.HaveReceivedVoltage1	Boolean		0
AlarmOnStartTime	Zones.1.AlarmOnStartTime	Integer		71
AlarmOnStartTime_0	Zones.1.AlarmOnStartTime.0	Integer		-1
AlarmOnStartTime_1	Zones.1.AlarmOnStartTime.1	Integer		-1
AlarmOnStartTime_2	Zones.1.AlarmOnStartTime.2	Integer		1

OK

Cancel

Update

Refresh

Header

Help

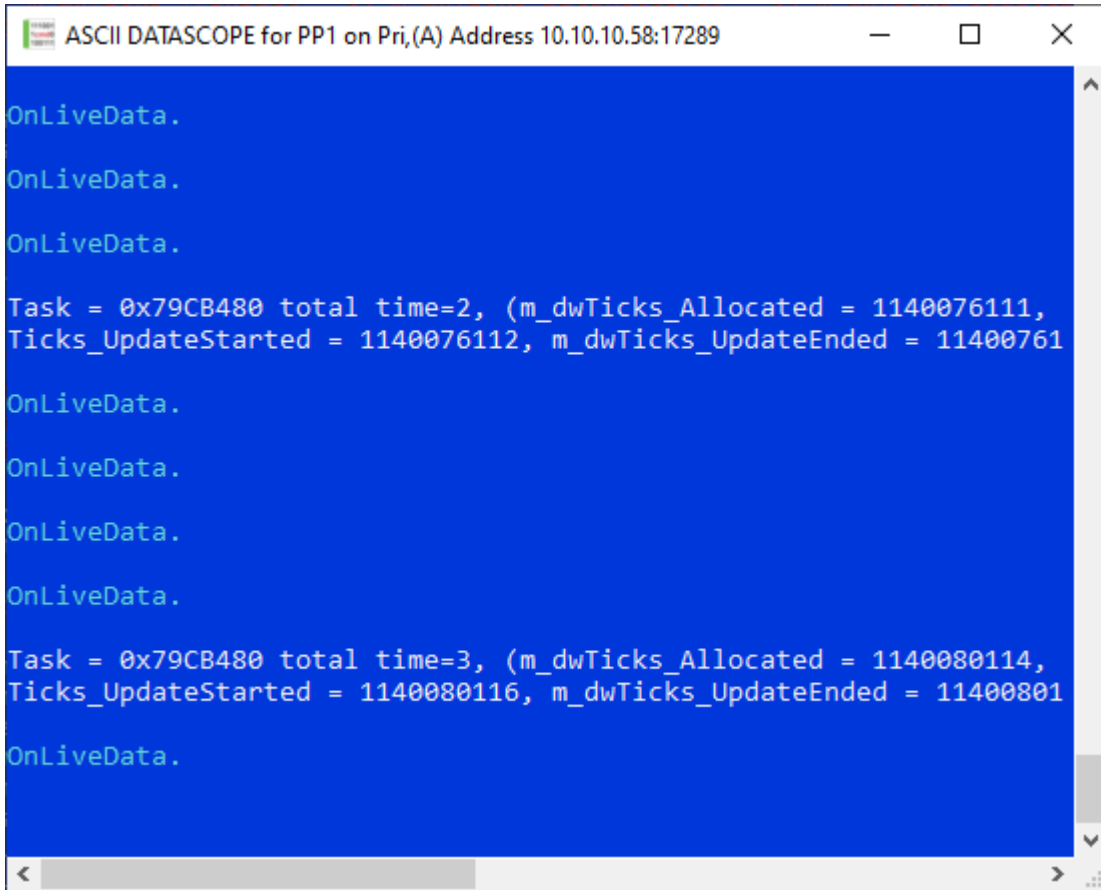
Load template

Save template

Expression:

5. Protocol Definition and Considerations

The following screen shot shows an example of the datascope under normal started conditions. Note the connection (primary) , IP address and port on the title bar:



```
ASCII DATASCOPE for PP1 on Pri,(A) Address 10.10.10.58:17289

OnLiveData.
OnLiveData.
OnLiveData.
Task = 0x79CB480 total time=2, (m_dwTicks_Allocated = 1140076111,
Ticks_UpdateStarted = 1140076112, m_dwTicks_UpdateEnded = 1140076113)
OnLiveData.
OnLiveData.
OnLiveData.
OnLiveData.
Task = 0x79CB480 total time=3, (m_dwTicks_Allocated = 1140080114,
Ticks_UpdateStarted = 1140080116, m_dwTicks_UpdateEnded = 1140080117)
OnLiveData.
```

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7. Revisions

7.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	28-Mar-2023	D. Jordaan	First release.	-
2.0				

7.2. DLL Revision

Rev.	Date	Changes
1	28/03/2023	First release of the driver.